Queenswood

WHY CHOOSE QUEENSWOOD

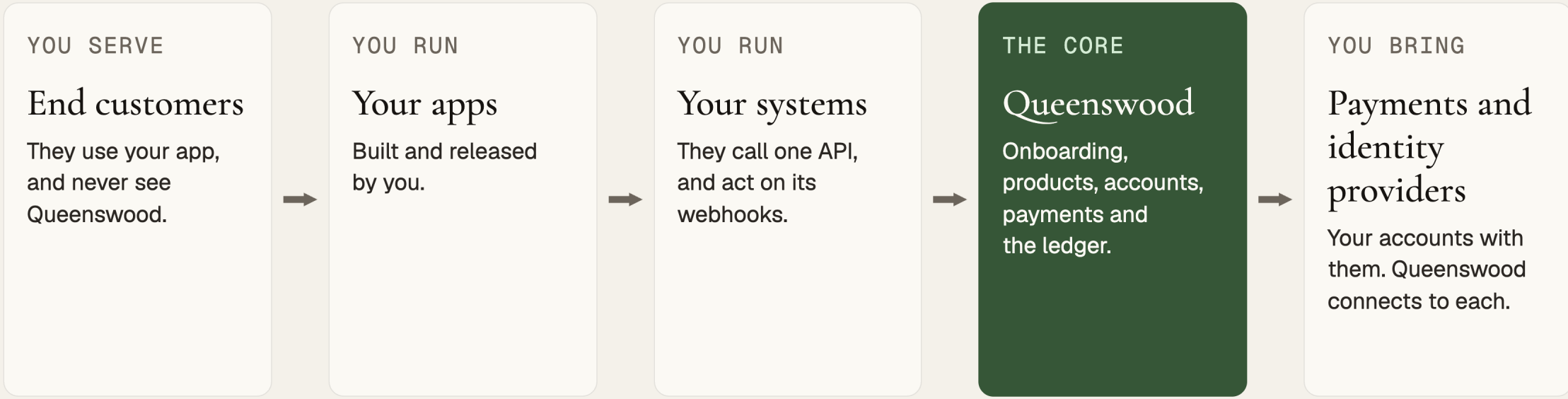
Open-source core banking.

For when you are modernising your banking platform, or embedding payments and accounts into your product.

— OVERVIEW

Where Queenswood sits

You still run your apps and your systems. Queenswood runs the banking behind them.



The picture is the same whether you are modernising your banking platform or embedding payments and accounts into your product.

THE PLATFORM

What Queenswood runs

All of it is configured and driven through one API.

Customer onboarding
Identity checks before any account.

Products and accounts
Versioned, published in the console.

Payments
UK Faster Payments, payee confirmed.

Interest and rewards
Accrued and paid on your schedule.

General ledger
Matching debits and credits.

Policies
Records checked on every request.

End-of-day processing
Scheduled, and every run recorded.

Webhooks
Tells your systems as things happen.

Operator console
Where your team runs the bank.

— 01 · OWNERSHIP

Open source

The code is yours to *read, run and change*. Contributions welcome.

COST

```
MIT License
Permission is hereby granted,
free of charge…
```

No licence fee, low running costs

Approx. £1,100/mo at today's Google Cloud list prices, as load tested.

RELEASES

v0.0.9	11 Oct 2026	latest
v0.0.8	7 Oct 2026	
v0.0.7	2 Oct 2026	

On your schedule

You decide when to take each release.

INFRASTRUCTURE



Where you choose

Your own infrastructure, or follow our Google Cloud blueprint.

— 02 · CHANGE

Configurable policies and products

Changing either *ships no code*.

POLICIES

DOMAIN	LIMITS	micro tier
Outbound Payment	MAX 500 count / day	
	MAX £10,000 GBP instant	
	MAX £50,000 GBP / day	

Rules are records, checked on every request

From simple limits to rules over rolling windows of activity.

PRODUCTS

Rainy Day holders onto v2 — completed

v1 4.10% → v2 4.35% published +0.25 pp

Rate schedules, fees and rewards

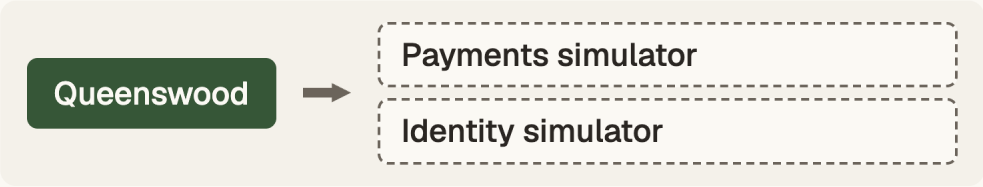
Published as versions, and migrated on your schedule. A new product line needs no code change.

— 03 · TESTING

Sandbox and pluggable providers

You build and test *before a contract is signed*.

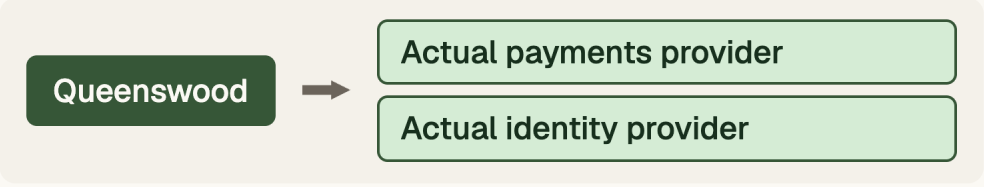
test



Your bank starts in test

Simulated money, on the same software your customers will use. No licence is needed.

live



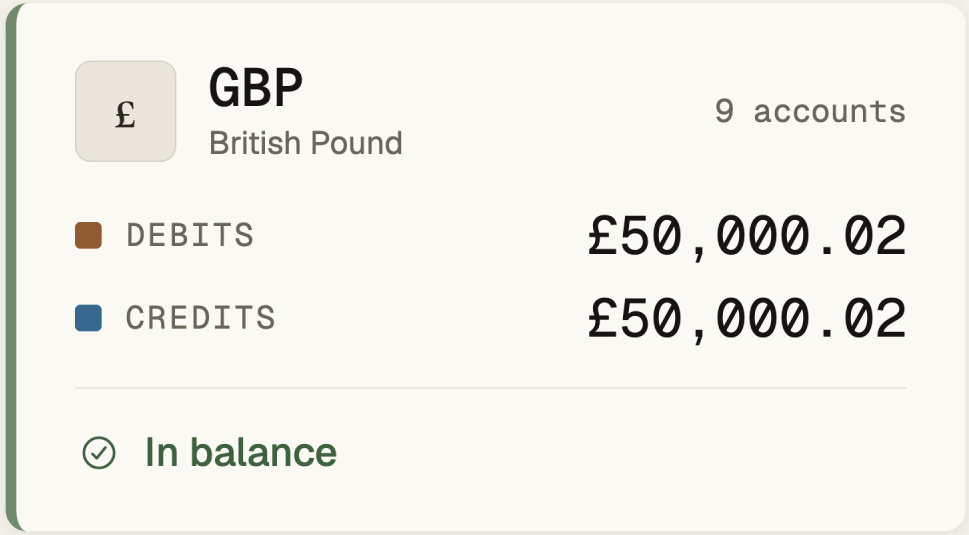
It moves to live when you are ready

Real money, through your own accounts with the providers you have contracted.

— 04 · THE LEDGER

Books that balance

Every movement is recorded as matching debits and credits.



GENERAL LEDGER	Your finance team reads it in the console.
TRIAL BALANCE	They check that it ties.
SUSPENSE	They see what waits in it.
INTEREST	It balances down to fractions of a penny.
UNDER LOAD	It balanced to the penny after every load-test run.

— 05 · CAPACITY

Measured under load, in order

A bank of 3 million customers makes 10 to 20 payments a second, and 3 to 5 times that at peak.

50

payments a second, steady

none failed · p50 25 ms · p99 50 ms

313

payments a second, at peak

p50 82 ms · p99 208 ms

0

payments refused at peak

each account's payments taken in the order accepted

3

8-vCPU machines, in a single zone

database at 2x replication, message bus at 3x

Release 0.0.9 on Google Cloud. Simulators stood in for the payments and identity providers, so these are the platform's figures, not a real provider's. Above its limit the API refused new requests at once and stayed up. No accepted payment was lost.

— 06 · ACCOUNTABILITY

Audit and access logs

Every change and every call names *who made it*: a person on your team, or your own systems.

YOUR TEAM `owner` `admin` `developer` `viewer` YOUR SYSTEMS `bank credential`

AUDIT LOG · FOR COMPLIANCE

13:53	Outbound payment submitted	Xepha	bank
13:52	Migration approved	Xepha Owner	
13:52	Product version published	Xepha Owner	

Every change to your bank's records

Who made it, when and what it changed. Nothing in it can be edited or deleted.

ACCESS LOG · FOR SECURITY

13:53	POST	SubmitOutboundPayment	201 replay
13:52	GET	RetrieveOutboundPayment	200
13:52	POST	SubmitInternalPayment	429 refused

Every authenticated call to the API

Who called, which operation, what came back and how long it took, refusals included.

— 07 · EVIDENCE

Documentation and tests

All of it says *what is done and what is not*.

DOCUMENTATION

PRDs requirements

TDDs designs

ADRs decisions

Recipes procedures

Written down, beside the code

Every requirement, design, decision and procedure.

TEST SUITE

MODEL = SYSTEM

no shared code

Checked against a model

Built separately. Same commands in, and the answers must agree.

COMPLIANCE

Data recovery → recipe

DORA CIS GDPR NIS2 ISO 22301

A register of obligations

Each links to the procedure that meets it, gaps included.

No accreditation is held yet.

PREREQUISITES

What you bring

1 A banking licence

Or a partner who holds one.

2 A payments provider

Your account with one. We connect to it for you.

3 An identity provider

Your account with one. Same again.

4 Somewhere to run it

Your own infrastructure, or follow our Google Cloud blueprint.

Bring nothing while you test.

The sandbox simulates your choice of payments and identity providers, and no licence is needed for simulated money.

NEXT STEP

Two ways to run it

Your bank starts in test, with simulated money.

ONLINE

Open a sandbox bank

queenswood.io →

Register, name your bank and choose its providers. Then run the scenarios in its console.

ON YOUR OWN INFRASTRUCTURE

Run it yourself

github.com/repldriven/queenswood →

Follow the readme to run it on a laptop or your own cluster, simulators included.

Read the source

In the same repository: demo videos, the API reference and design documents.

— HOW IT IS BUILT

Technology stack

API	OpenAPI 3.x, OAuth 2.0, OpenID Connect	One API, documented from its own routes.
SERVICES	Clojure on the JVM	Separate services, or one process.
MESSAGING	Kafka or Pulsar, with Avro	Commands and events, kept in order.
DATABASE	FoundationDB Record Layer	All-or-nothing writes across records.
OBSERVABILITY	OpenTelemetry and SigNoz	Request tracing and performance metrics.
DEPLOYMENT	Kubernetes and Helm	One chart, onto any Kubernetes cluster.
CLOUD	Crossplane and Argo CD on Google Cloud	Every change is a reviewed pull request.
DEVELOPMENT	AI-assisted	Written docs auto-sync to agent skills.

WHY QUEENSWOOD

Our mission

To put the software that runs a bank *in the open*, for anyone to read, run and improve.

Open to review

Your security team, auditors and regulator can read the code that moves the money, and run the load tests that prove it works.

Yours to keep

Whatever happens to us, you keep the code you run and the right to change it.

Open to contribution

Contributors can add products, providers and new capabilities, and fix faults. Every bank that runs it gets them all.

Path to production

Most open-source projects stop at the code. Ours has a Google Cloud blueprint to deploy, scale and run as many instances as you need.